



**PHILOMATH
NETWORKS PVT. LTD**



**6-Days Hands-On
Workshop
on**

**"DevOps :
Cloud, Automation & CI/CD
Mastery"**

**Get ready to earn your DevOps
Certification with us!**

**Redefining Learning
Beyond Boundaries**



**Phone Number
+91 88698 32559**

About Philomath Networks



At Philomath Networks Pvt. Ltd., we specialize in organizing transformative events for schools, organizations, and communities, fostering a culture of exploration, innovation, and growth. From expert-led workshops to competitive events, we create immersive learning experiences that inspire and empower individuals to go beyond boundaries. Partner with us to bring impactful learning to your community!



Partner With Us

Collaborate with Philomath Networks to bring innovation, engagement, and excellence to your institution or organization. Let's work together to create an ecosystem of lifelong learning and impactful experiences!

What Sets Us Apart?

Comprehensive Learning Approach – We blend academic, non-academic, and skill-based experiences through workshops, competitions, expert-led sessions, and immersive programs tailored to real-world applications.

End-to-End Event Management – From conceptualization to execution, we ensure seamless organization of competitions, tech fairs, hackathons, leadership programs, and customized training sessions that align with institutional goals.

Industry & Academic Bridging – We foster collaborations between institutions, corporates, and experts, ensuring that participants gain practical exposure, networking opportunities, and insights into emerging trends beyond traditional education.

Customizable & Scalable Programs – Whether it's a school-level science fest, university hackathon, corporate leadership training, or specialized certification courses, we tailor our events to meet the unique needs of every institution.

Proven Impact & Engagement – Our interactive approach ensures higher participation, deeper learning retention, and enhanced skill development, making every event a transformative experience.

Course Overview



Course Objective

The DevOps Pro: Cloud, Automation & CI/CD Mastery Certification provides a strong foundation in DevOps principles, automation, and cloud-native development. Learners will gain hands-on experience with CI/CD pipelines, Docker, Kubernetes, Terraform, and cloud deployments. They will develop skills in infrastructure as code, monitoring, and security best practices (DevSecOps). By the end of the course, they will be equipped to streamline software delivery, manage scalable infrastructure, and optimize deployment workflows, making them industry-ready DevOps professionals.

Course Outcome

✓ Master DevOps workflows and automation tools.

✓ Automate infrastructure with Terraform and Ansible.

✓ Implement CI/CD pipelines for efficient software delivery.

✓ Enhance security with DevSecOps best practices.

✓ Deploy and manage containers using Docker and Kubernetes.

✓ Optimize cloud deployments on AWS, Azure, or GCP.

Training

Modules

The training modules cover core DevOps concepts, automation, and cloud-native development through hands-on learning. Learners will start with version control, CI/CD pipelines, and configuration management, gaining expertise in tools like Git, Jenkins, and Ansible. They will then dive into containerization with Docker and orchestration with Kubernetes, enabling scalable and efficient deployments. Each module is designed to provide practical experience through real-world scenarios, ensuring a deep understanding of DevOps tools and methodologies. Participants will also learn best practices to enhance collaboration and streamline development workflows.

Module 1: DevOps Foundations & Version Control

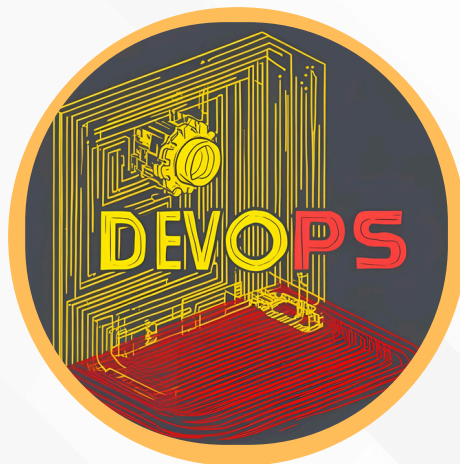
- Understand DevOps principles, lifecycle, and key practices.
- Learn the differences between DevOps, Agile, and traditional IT.
- Get familiar with essential DevOps tools and technologies.
- Master version control with Git and repository management.
- Learn branching, merging, and collaboration with GitHub.
- Gain hands-on experience with Git workflows and best practices.

Module 2: CI/CD Pipelines & Configuration Management

- Explore Continuous Integration and Continuous Deployment (CI/CD).
- Set up Jenkins for automated build and deployment pipelines.
- Learn Jenkins pipeline scripting and job automation.
- Understand Ansible for configuration management.
- Write and execute Ansible playbooks for system automation.
- Implement real-world CI/CD pipelines with hands-on practice.

Module 3: Containerization with Docker

- Understand containerization concepts and benefits.
- Install and configure Docker for development and deployment.
- Manage Docker images, containers, and networking.
- Use Docker Compose to manage multi-container applications.
- Learn Docker storage, volumes, and persistent data handling.
- Deploy applications using Docker best practices.



Training Modules

As the course progresses, participants will explore infrastructure automation with Terraform, monitoring with Prometheus and Grafana, and security best practices with DevSecOps. The final modules focus on cloud-based DevOps workflows, secrets management, and compliance, ensuring learners are industry-ready to implement reliable, scalable, and secure DevOps solutions. By the end of the training, participants will have the confidence to design, implement, and manage DevOps workflows in real-world environments. This comprehensive approach will prepare them for DevOps roles in leading tech industries.

Module 4: Kubernetes & Orchestration

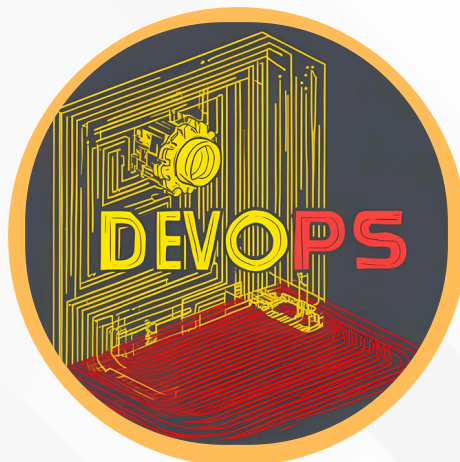
- Learn Kubernetes architecture, components, and cluster setup.
- Deploy and manage applications with Kubernetes pods and services.
- Implement Kubernetes networking, ingress, and service discovery.
- Use ConfigMaps and Secrets for application configuration.
- Understand scalability, load balancing, and auto-healing.
- Gain hands-on experience with real-world Kubernetes deployments

Module 5: Infrastructure as Code & Monitoring

- Learn the fundamentals of Infrastructure as Code (IaC).
- Automate cloud infrastructure using Terraform.
- Deploy and manage resources efficiently with Terraform scripts.
- Set up monitoring with Prometheus and Grafana.
- Implement centralized logging using the ELK stack.
- Gain hands-on experience in automating infrastructure and monitoring

Module 6: Cloud DevOps & Security Best Practices

- Explore cloud-based DevOps workflows (AWS, Azure, GCP).
- Deploy CI/CD pipelines and applications on cloud platforms.
- Implement DevSecOps security best practices.
- Learn secrets management with Vault and AWS Secrets Manager.
- Understand compliance, governance, and security policies.
- Work on a capstone project integrating all DevOps concepts.



Training Schedule & Format



The training schedule is structured as a 6-day intensive DevOps workshop, providing a hands-on learning experience. Each day focuses on a key DevOps module, starting with foundational concepts and progressing to advanced topics like CI/CD pipelines, containerization, orchestration, and cloud automation. The workshop includes interactive sessions, hands-on labs, and real-world DevOps problem-solving to reinforce learning. Participants will work on mini-projects and practical assignments daily, ensuring a strong grasp of tools and techniques. The final day is dedicated to a capstone project, workflow optimization, and a certification assessment, equipping learners with industry-ready DevOps skills.

Day	Module	Focus Area	Duration
Day 1	DevOps Foundations & Version Control	Concepts, Git, GitHub	5hrs
Day 2	CI/CD Pipelines & Automation	Jenkins, Ansible, Scripting	5hrs
Day 3	Containerization with Docker	Images, Containers, Compose	5hrs
Day 4	Kubernetes & Orchestration	Clusters, Deployments, Scaling	5hrs
Day 5	Infrastructure as Code & Monitoring	Terraform, Prometheus, ELK	5hrs
Day 6	Cloud DevOps & Security	AWS, DevSecOps, Compliance	5hrs

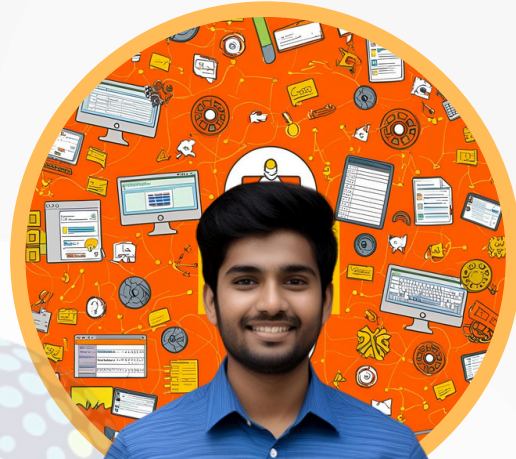
Through hands-on exercises and case studies, participants will develop essential DevOps skills, explore best practices, and tackle real-world automation challenges. The workshop includes live demonstrations, guided projects, and interactive labs to enhance learning. A collaborative format encourages discussions, troubleshooting, and peer learning. By the end, attendees will have a strong DevOps foundation, enabling them to automate workflows, optimize deployments, and manage scalable infrastructure effectively.

How to Get Involved



Steps to Enroll in Training Programs

- Contact us via email or phone.
- Share your preferred course, schedule, and any specific requirements.
- Our team will review your request and confirm the booking.
- You will receive confirmation details via email or call .



Contact Information

For any questions about the training program, please reach out to:

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